Robert Schweller

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Brief Biography

Robert T. Schweller is a Ph.D. student in Computer Science at Northwestern University in Evanston. His research interests are in the design and analysis of algorithms. Emphases include DNA self assembly, combinatorial DNA code word design, and anomaly/intrusion detection over network data streams.

Education

Northwestern University, Evanston, IL

Master of Science in Computer Science, September 2002 - June 2004.
Course topics include Advanced Algorithms, Bioinformatics, Behavior Based Robotics, Databases, Game Theory.
Ph.D. Dissertation Submitted, January 2007
Ph.D. in Computer Science, (anticipated) June 2007

Carleton College, Northfield, MN

Bachelor of Arts in Mathematics, June 2001.
Advanced course topics in Computer Science and Mathematics include Abstract Algebra, Combinatorics, Complex Analysis, Computer Graphics, Measure Theory, Programming Languages.
Graduated Cum Laude.

Teaching and Mentorship

Teaching and Tutoring

I have served as the teaching assistant or tutored for the following courses. Teaching Assistant duties include grading homework and exams, holding office hours, holding supplementary review sessions, and periodically providing core lectures. Tutorship duties include assisting groups and individuals with various subject material.

- **(TA)** Mathematical Foundations of Computer Science, Northwestern University — 2006
- **(TA)** Introduction to Algorithms, Northwestern University — four terms, 2002 – 2006
- **(TA)** Introduction to Computer Science, Carleton College — 2001
- **(Tutor)** I have served as a Carleton College peer tutor for courses in Calculus I-III, Linear Algebra, and Discrete Mathematics. — 1998-2000

Advising and Mentoring

- Coach for two Northwestern University programming teams competing in the ACM International Collegiate Programming Contest — November, 2006
- Supervised the research of a Northwestern graduate student working on a project within my domain of expertise. — 2006-

Referee Work

- Reviewer, Siam Journal on Computing, 2006
- Reviewer, IEEE Transactions on Nanobioscience, 2006
- Reviewer, Conference on Foundations of Software Technology and Theoretical Computer Science, 2007
- Reviewer, ACM-SIAM Symposium on Discrete Algorithms, 2007
- Reviewer, ACM Symposium on Theory of Computing, 2007
- Reviewer, Theoretical Computer Science, 2007

Publications/Patents:

Patents:

1. **Reversible Sketch**
   Patent filed, March 2006.

Dissertation:

2. **Complexities for the Design of Self-Assembly Systems**
   Ph.D. dissertation, Northwestern University, submitted January 2007.

Journal Papers:

3. **Complexities for Generalized Models of Self-Assembly**
   Gagan Aggarwal, Qi Cheng, Michael H. Goldwasser, Ming-Yang Kao, Pablo Moisset de Espanes, Robert T. Schweller
   SIAM Journal on Computing (SICOMP), Volume 34, Number 6, pp. 1493-1515.
4. **Reversible Sketches: Enabling Monitoring and Analysis over High-speed Data Streams**
   Robert Schweller, Zhichun Li, Yan Chen, Yan Gao, Ashish Gupta, Yin Zhang, Peter Dinda, Ming-Yang Kao, Gokhan Memik
   Accepted to IEEE/ACM Transactions on Networking.
5. **HRAID: a High-speed Router based Anomaly/Intrusion Detection System**
   Yan Gao, Zhichun Li, Robert Schweller, Yan Chen
   Under submission.
6. **Randomized Fast Design of Short DNA Words**
   Ming-Yang Kao, Manan Sanghi, Robert Schweller
   Under submission.
7. **Reducing Tile Complexity for Self-Assembly Through Temperature Programming**
   Ming-Yang Kao, Robert Schweller
   Under submission.
8. **Flexible Word Design and Graph Labeling**
   Ming-Yang Kao, Manan Sanghi, Robert Schweller
   Under submission.

Conference Papers:

9. **Flexible Word Design and Graph Labeling**
   Ming-Yang Kao, Manan Sanghi, Robert Schweller
   Proceedings of the 17th International Symposium on Algorithms and Computation (ISAAC 2006), Kolkata, India, Dec. 2006, pp. 48-60.
10. **Reverse Hashing for High-speed Network Monitoring: Algorithms, Evaluation, and Applications**
    Robert Schweller, Zhichun Li, Yan Chen, Yan Gao, Ashish Gupta, Yin Zhang, Peter Dinda, Ming-Yang Kao, Gokhan Memik
    Proceedings of the 25th Annual Joint Conference of the IEEE Computer and Communications Societies (Infocom 2006), Barcelona, Spain, April 2006.
11. **Reducing Tile Complexity for Self-Assembly Through Temperature Programming**
    Ming-Yang Kao, Robert Schweller
    Proceedings of the 17th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2006), Miami, Florida, Jan. 2006, pp. 571-580.

12. **Randomized Fast Design of Short DNA Words**
    Ming-Yang Kao, Manan Sanghi, Robert Schweller
    Proceedings of the 32nd International Colloquium on Automata, Languages and Programming (ICALP 2005), Lisboa, Portugal, July 11-15, 2005, pp. 1275-1286.
13. **Reversible Sketches for Efficient and Accurate Change Detection over Network Data Streams**
    Robert Schweller, Ashish Gupta, Elliot Parsons, Yan Chen
    Proceedings of ACM SIGCOMM Internet Measurement Conference (IMC 2004), Taormina, Sicily, Italy, October 25-27, 2004, pp. 207-212.
14. **Complexities for Generalized Models of Self-Assembly**
    Gagan Aggarwal, Michael H. Goldwasser, Ming-Yang Kao, Robert T. Schweller
    Proceedings of the 15th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2004), New Orleans, Louisiana, Jan. 2004, pp. 880-889.
15. **Randomized Self-Assembly for Approximate Shapes**
    Ming-Yang Kao, Robert Schweller
    Under submission.
16. **Staged Self-Assembly: Nanomanufacture of Arbitrary Shapes with O(1) Glues**
    Erik D. Demaine, Martin L. Demaine, Sandor P. Fekete, Mashhood Ishaque, Eynat Rafalin, Robert Schweller, Diane L. Souvaine
    Under submission.

Presentations

- **DePaul Theory Seminar** – Chicago, IL — May 12, 2006
  Seminar Talk. *Complexities for the Design of Self-Assembly Systems*.
- **MIT – CSAIL Theory Seminar** – Cambridge, MA — May 3, 2006
  Seminar Talk. *Complexities for the Design of Self-Assembly Systems*..
- **SODA 2006** – Miami, FL — January 23, 2006
  Conference Talk. *Reducing Tile Complexity for Self-Assembly Through Temperature Programming*.
- **Midwest Theory Day** – Milwaukee, WI — December 10, 2005
  Workshop Talk. *Reducing Tile Complexity for Self-Assembly Through Temperature Programming*.
- **Buena Vista University Speaker Series 'Speaking of Science'** - Storm Lake, IA — February 28, 2005
  Invited Talk. *DNA Self-Assembly*
- **Midwest Theory Day** – Chicago, IL — December 4, 2004
  Workshop Talk. *Reversible Sketches for Efficient and Accurate Change Detection over Network Data Streams*.
- **IMC 2004** - Taormina, Sicily, Italy — October 26, 2004
  Conference Talk. *Reversible Sketches for Efficient and Accurate Change Detection over Network Data Streams*.
- **SODA 2004** – New Orleans, Louisiana — January 13, 2004
  Conference Talk. *Complexities for Generalized Models of Self-Assembly*.
- **Midwest Theory Day** – Chicago, IL — December 13, 2003
  Workshop Talk. *Complexities for Generalized Models of Self-Assembly*.

References

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Yan Chen
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